THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME — SET A

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** potassium	Alkali metal
2	**(c)** bromine	Mercury is a metal
3	**(b)** silver and copper	Lead and mercury are poor
4	**(c)** Incorrect — mercury is a liquid metal	Mercury
5	**(a)** Student A is correct	Reacts with both acids and bases
6	**(d)** silver sulphide	H_2S in air
7	**(d)** sodium	Catches fire in air
8	**(c)** galvanisation	Zinc protects iron
9	**(c)** 3 : 1	Dissolves gold
10	**(c)** high melting points	Strong electrostatic forces

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It reacts vigorously with oxygen and moisture and can catch fire; kerosene prevents contact with air.	2
12	Metal: mercury; non-metal: bromine	1+1
13	A homogeneous mixture of a metal with other metals or a non-metal; e.g. brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn)	1+1
14	False — HNO_3 is a strong oxidising agent; it oxidises H_2 to water (except Mg and Mn with very dilute HNO_3).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$Na \longrightarrow Na^+ + e^-$, $Cl + e^- \longrightarrow Cl^-$ ⇒ Na^+Cl^-; $Mg \longrightarrow Mg^{2+} + 2e^-$, two Cl atoms accept one electron each ⇒ $Mg^{2+}(Cl-)_2$	1½+1½
16	Oxides reacting with both acids and bases: Al_2O_3, ZnO. $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$; $Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$	1+1+1
17	(i) $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (ii) $Ca + 2\,H_2O \longrightarrow Ca(OH)_2 + H_2$ (iii) $2\,K + 2\,H_2O \longrightarrow 2\,KOH + H_2$ + heat	1+1+1
	OR: (i) lustrous, unreactive, do not corrode (ii) highly reactive with air and water (iii) light, good conductor of heat, resistant to corrosion due to oxide layer	1+1+1
18	Reaction of iron(III) oxide with aluminium: $Fe_2O_3 + 2\,Al \longrightarrow 2\,Fe + Al_2O_3$ + heat; used to join railway tracks/cracked machine parts	1+1+1
	OR: (i) ions become free to move (ii) strong inter-ionic attraction needs a lot of energy (iii) water is a polar solvent; kerosene is non-polar	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) metals arranged in decreasing order of reactivity (ii) roasting: $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$; reduction: $ZnO + C \longrightarrow Zn + CO$ (iii) it is easier to obtain a metal from its oxide	1+3+1
	OR: (i) purifying an impure metal by electrolysis (ii) impure Cu as anode, pure Cu strip as cathode in acidified $CuSO_4$ solution (iii) anode — impure Cu, cathode — pure Cu, electrolyte — acidified $CuSO_4$ (iv) below the anode	1+2+1+1
20	(i) X: gold/silver; Y: magnesium; Z: sodium (ii) $2\,Mg + O_2 \longrightarrow 2\,MgO$ (iii) Z has more affinity for oxygen than carbon	3+1+1
	OR: (i) (a) metals mostly solids; non-metals solids/liquid/gases (b) metals lustrous, non-metals dull (c) metals good conductors, non-metals poor (d) metal oxides basic, non-metal oxides acidic/neutral (ii) iodine; graphite	4+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** malleability	Gold and silver are most malleable
2	**(d)** lead	Lead and mercury
3	**(b)** sodium	Very reactive
4	**(b)** Incorrect — graphite conducts electricity	Exception: graphite
5	**(a)** Student B is correct	Acidic oxides are of non-metals
6	**(d)** cinnabar (HgS)	Roasted to give Hg
7	**(d)** zinc	Zinc coating
8	**(d)** zinc	Zn is more reactive
9	**(c)** bronze	Brass is Cu + Zn
10	**(a)** transfer of electrons	Electrovalent bond

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bubbles of hydrogen gas formed stick to the surface of calcium.	2
12	Gold, silver, platinum (any two)	1+1
13	Minerals: naturally occurring compounds of metals; ores: minerals from which a metal can be extracted profitably.	2
14	False — 24 carat gold is very soft; it is alloyed with copper or silver (22 carat) to make it harder.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Na (2,8,1): one dot; O (2,6): six dots; Mg (2,8,2): two dots. $Mg \longrightarrow Mg^{2+} + 2\,e^-$, $O + 2\,e^- \longrightarrow O^{2-} \Rightarrow Mg^{2+}\ O^{2-}$	1+2
16	(i) $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (catches fire) (ii) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (iii) $2\,Cu + O_2 \longrightarrow 2\,CuO$	1+1+1
17	Slow eating of metal by air/moisture; prevented by painting, oiling, galvanising, alloying; zinc protects iron even if the coating is broken.	1+1+1
	OR: (i) they are very reactive; carbon cannot reduce their oxides (ii) copper is less reactive than hydrogen (iii) sodium has greater affinity for oxygen than carbon	1+1+1
18	Roasting: heating sulphide ore in air, $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$. Calcination: heating carbonate ore in limited air, $ZnCO_3 \longrightarrow ZnO + CO_2$.	1½+1½
	OR: (i) aircraft — light, strong (ii) electric wires — good conductor, ductile (iii) machinery/construction — strong (as steel)	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$. $2\,Cu_2S + 3\,O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2$; $2\,Cu_2O + Cu_2S \longrightarrow 6\,Cu + SO_2$ (ii) they are low in the reactivity series	4+1
	OR: (i) homogeneous mixtures of metals with metals/non-metals (ii) to increase hardness, strength, resistance to corrosion; lower melting point (iii) brass: Cu + Zn; bronze: Cu + Sn; solder: Pb + Sn (iv) welding electrical wires	1+1+2+1
20	(i) D (displaces Fe) (ii) A or C (most likely A, as it cannot displace Fe) (iii) D > B > A (iv) copper	1+1+2+1
	OR: (i) compounds formed by transfer of electrons from metal to non-metal (ii) solids, high melting/boiling points, soluble in water, conduct in molten/solution state (iii) they contain oppositely charged ions held by electrostatic forces	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.